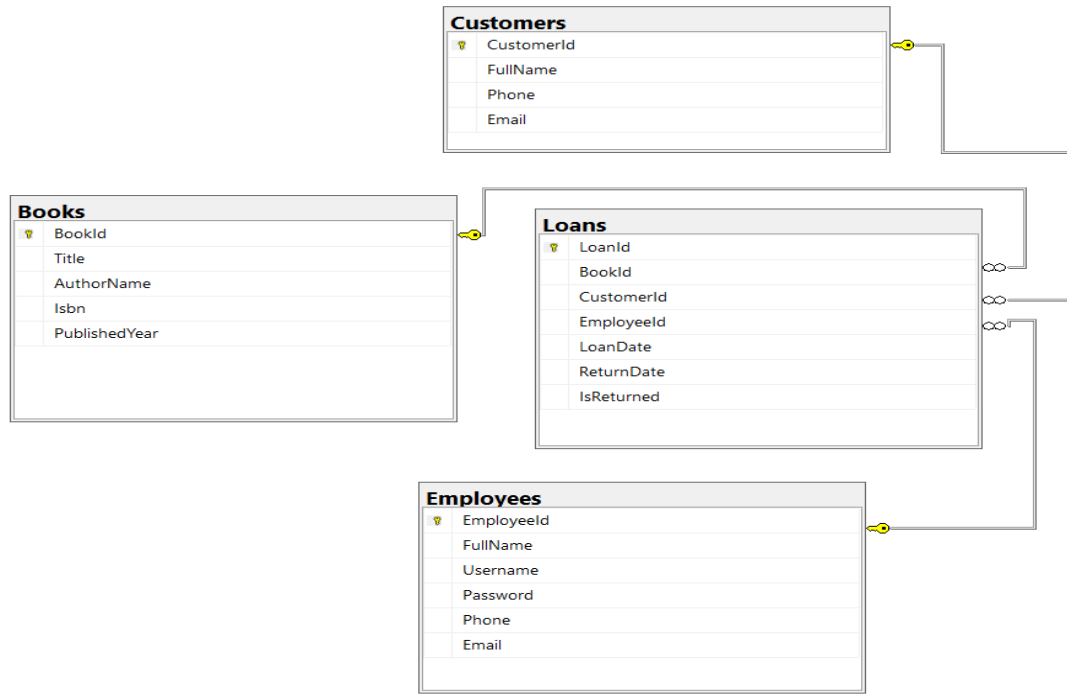


Library Management System- Database Design



Results

Messages

	EmployeeId	FullName	Username	Password	Phone	Email
1	1	Aisha Ali	aisha.admin	pass123	0612345678	aisha@library.com
2	2	Omar Hassan	omar.lib	lib456	0618765432	omar@library.com

	CustomerId	FullName	Phone	Email
1	1	Farah Abdi	0611112222	farah.abdi@email.com
2	2	Zahra Hussein	0613334444	zahra.h@email.com
3	3	Khalid Mohamed	0615556666	khalid.m@email.com

	BookId	Title	AuthorName	Isbn	PublishedYear
1	1	Clean Code	Robert C. Martin	978-0132350884	2008
2	2	The Pragmatic Programmer	Andrew Hunt	978-0135957059	1999
3	3	Introduction to Algorithms	Thomas H. Cormen	978-0262033848	2009

	LoanId	BookId	CustomerId	EmployeeId	LoanDate	ReturnDate	IsReturned
1	1	1	1	1	2026-06-10	NULL	0
2	2	2	2	2	2026-06-12	2026-06-15	1
3	3	3	3	1	2026-06-16	NULL	0

```
CREATE DATABASE LibraryManagementSystem;
```

```
GO
```

```
USE LibraryManagementSystem;
```

```
GO
```

```
CREATE TABLE Employees (
```

```
    EmployeeId INT IDENTITY(1,1) PRIMARY KEY,
```

```
    FullName VARCHAR(100) NOT NULL,
```

```
    Username VARCHAR(50) NOT NULL,
```

```
    Password VARCHAR(255) NOT NULL,
```

```
    Phone VARCHAR(20),
```

```
    Email VARCHAR(100)
```

```
);
```

```
CREATE TABLE Customers (
```

```
    CustomerId INT IDENTITY(1,1) PRIMARY KEY,
```

```
    FullName VARCHAR(100) NOT NULL,
```

```
    Phone VARCHAR(20),
```

```
    Email VARCHAR(100)
```

```
);
```

```
CREATE TABLE Books (
```

```
    BookId INT IDENTITY(1,1) PRIMARY KEY,
```

```
Title VARCHAR(200) NOT NULL,  
AuthorName VARCHAR(100) NOT NULL,  
Isbn VARCHAR(50),  
PublishedYear INT  
);
```

```
CREATE TABLE Loans (  
    LoanId INT IDENTITY(1,1) PRIMARY KEY,  
    BookId INT NOT NULL FOREIGN KEY REFERENCES Books(BookId),  
    CustomerId INT NOT NULL FOREIGN KEY REFERENCES Customers(CustomerId),  
    EmployeeId INT NOT NULL FOREIGN KEY REFERENCES Employees(EmployeeId),  
    LoanDate DATE NOT NULL,  
    ReturnDate DATE NULL,  
    IsReturned BIT DEFAULT 0  
);  
GO
```

```
INSERT INTO Employees (FullName, Username, Password, Phone, Email)  
VALUES  
('Aisha Ali', 'aisha.admin', 'pass123', '0612345678', 'aisha@library.com'),  
('Omar Hassan', 'omar.lib', 'lib456', '0618765432', 'omar@library.com');
```

```
INSERT INTO Customers (FullName, Phone, Email)  
VALUES  
('Farah Abdi', '0611112222', 'farah.abdi@email.com'),  
('Zahra Hussein', '0613334444', 'zahra.h@email.com'),
```

```
('Khalid Mohamed', '0615556666', 'khalid.m@email.com');
```

```
INSERT INTO Books (Title, AuthorName, Isbn, PublishedYear)
```

```
VALUES
```

```
('Clean Code', 'Robert C. Martin', '978-0132350884', 2008),
```

```
('The Pragmatic Programmer', 'Andrew Hunt', '978-0135957059', 1999),
```

```
('Introduction to Algorithms', 'Thomas H. Cormen', '978-0262033848', 2009);
```

```
INSERT INTO Loans (BookId, CustomerId, EmployeeId, LoanDate, ReturnDate, IsReturned)
```

```
VALUES
```

```
(1, 1, 1, '2026-06-10', NULL, 0),
```

```
(2, 2, 2, '2026-06-12', '2026-06-15', 1),
```

```
(3, 3, 1, '2026-06-16', NULL, 0);
```

```
GO
```